

Optimization

Exercise 1

Consider a transmitter at a location $\mathbf{x}_t = (-2, -4)$. It is required to find the optimal location to place a receiver $\mathbf{x}_r$ within a specified area of land such that the received signal's power is maximized. The received signal's power is inversely proportional to the squared distance between the transmitter and the receiver:

$$P(\mathbf{x}) \propto \frac{1}{d^2} \quad (1)$$

The allowable area to place the receiver is given by the following constraints in the euclidean plane:

$$c_1(\mathbf{x}) = -x_1^2 - (x_2 + 4)^2 + 16 \geq 0 \quad (2)$$

$$c_2(\mathbf{x}) = x_1 - x_2 - 6 \geq 0 \quad (3)$$

(a) Sketch the feasible region.

(b) Formulate the optimization problem (Write-up a cost function with the constraints).

■ *Hint:* Maximizing the power is equivalent to minimizing the distance. It will make your cost function easier to deal with.

(c) Solve the problem graphically.

■ *Hint:* Sketch the contour levels of the objective function.

(d) Verify that your solution satisfies the first and second-order necessary conditions for optimality.

■ *Hint:* Remember to check for feasible directions if needed.

(e) What is the worst location for the receiver to be?

(f) Is the solution to the constrained problem the same as the solution to the unconstrained problem?

(g) Consider now an additional constraint defined as:

$$c_3(\mathbf{x}) = x_1^2 + (x_2 + 6)^2 \geq 2. \quad (4)$$

Re-sketch the feasible region.

(h) Solve the new problem graphically.